

Sparse pathways for interpretable language models

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ABSTRACT

We investigate sparse activation pathways in compact language models. This fictional manuscript demonstrates private review and annotations. The example study compares two synthetic benchmarks. No real research result is claimed, and this document should not be cited as evidence.

1. RESEARCH QUESTION

Can a constrained activation pathway help a reader understand which features influence a model output? Interpretability is treated as a measurable research question, rather than a guarantee of correctness.

2. PROPOSED EVALUATION

Compare a sparse pathway with a dense baseline under matched training conditions. Report uncertainty across seeds, ablate the sparsity budget, and test whether results transfer beyond the synthetic benchmarks.

3. LIMITATIONS AND OPEN QUESTIONS

Two synthetic benchmarks cannot establish general performance. External evaluation, stronger baselines, and human assessment remain necessary. Correlation between sparsity and interpretation does not establish a causal relationship.

Select a sentence to highlight it and add a private or shared note.